

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re: U.S. Patent No. 7,504,095 Issue Date: March 17, 2009
Serial No: 11/343,489 Filing or 371(c) Date: January 31, 2006
Inventors: SCHWARZ et al. Docket No. 074540-0568897
Title: **METHOD FOR OBTAINING PRECURSOR Z AND USE THEREOF
FOR THE PRODUCTION OF A MEANS FOR THERAPY OF HUMAN
MOLYBDENUM COFACTOR DEFICIENCY**

Mail Stop Hatch-Waxman PTE
Office of Patent Legal Administration
Room MDW 7D55
600 Dulany Street (Madison Building)
Alexandria, VA 22314

**COVER LETTER FOR SUPPLEMENT TO APPLICATION OF
PATENT TERM UNDER 35 U.S.C. §156**

Dear Madam or Sir:

Further in connection with the pending Application for Extension of Patent Term under 35 U.S.C. §156 of U.S. Patent No. 7,504,095, filed April 16, 2021, Applicant respectfully requests the following amendments thereto (the “Supplemental”). This paper is Supplemental to the Application for Extension of Patent Term under 35 U.S.C. §156 of U.S. Patent No. 7,504,095 filed April 16, 2021. By this Supplemental, the Applicant information is updated in the Introductory Comments, and in applicable paragraphs in Section 11, Footnote 6, Section 14, and Section 15. Exhibit 2 is replaced by Substitute Exhibit 2, and Exhibits 22-24 are added. According, upon entry of the Supplemental paper, all references to Applicant in the Application for Extension of Patent Term under 35 U.S.C. §156 of U.S. Patent No. 7,504,095 filed April 16, 2021, shall mean Sentyln Therapeutics, Inc.

Applicant notes that this Supplemental paper does not restate the Application for Extension of Patent Term under 35 U.S.C. §156 of U.S. Patent No. 7,504,095 filed April 16, 2021, but updates the Application, with copies of newly recorded assignments, and updates the Name and Address for Correspondence.

The Supplement contains the following **Supplemental Exhibits**:

Substitute Exhibit 2: Abstract of Chain of Title from USPTO Assignment Database

Exhibit 22: Assignment Cover Sheet and Assignment from ORIGIN BIOSCIENCES, INC. to SENTYNL THERAPEUTICS, INC. (reel/frame 059671/0674)

Exhibit 23: March 31, 2022, letter from Sentynl Therapeutics, Inc. to FDA providing notice of the acquisition of full ownership of and all rights to NULIBRY NDA number 214018, attachments and acknowledgement of receipt from the FDA.

Exhibit 24: March 31, 2022, letter from Origin Biosciences, Inc. to FDA providing notice of the change of ownership of the NULIBRY NDA number 214018 to Sentynl Therapeutics, Inc. and acknowledgement of receipt from the FDA.

The Commissioner is hereby authorized to charge any fees that may be necessitated in connection with the filing of the Supplemental to Deposit Account No. 033975.

If there are any questions, Applicant's representative can be reached at 858-509-4071.

In accordance with the Notice of May 29, 2020, allowing for the initial filing of Application for Extension of Patent Term under 35 U.S.C. §156 via EFS-web and waiving the requirements under 37 C.F.R. 1.740(b) and 37 C.F.R. 1.790(b) for two additional copies of the Application, Applicant is submitting a single copy of this Supplement via EFS-web. As required by the Notice, contact information for the undersigned includes an email address.

Sincerely,

/Michelle L. Mehok/

Michelle L. Mehok
Reg. No. 53,314
Customer No. 27500
PILLSBURY WINTHROP SHAW PITTMAN LLP
11682 El Camino Real
Suite 200
San Diego, California 92130
Email: Docket_IP@pillsburylaw.com
Telephone: 858-509-4071
Facsimile: 858-509-4010

Date: August 1, 2022

Please replace the first paragraph in the INTRODUCTORY COMMENTS to read -

Sentynl Therapeutics, Inc. (“Sentynl”) is the successor of interest of Origin Biosciences, Inc. rights as the prior patent owner of United States Letters Patent No. 7,504,095 and Market Applicant for NULIBRY (fosdenopterin) New Drug Application (“NDA”) number 214018. On April 16, 2021, Origin Biosciences, Inc. timely submitted an Application for Extension of Patent Term under 35 U.S.C. §156 to extend the term of United States Letters Patent No. 7,504,095 based upon the regulatory review period that ended with the Food and Drug Administration’s (“FDA”) approval of NDA number 214018 for NULIBRY (fosdenopterin) for injection, for intravenous use. As the current patent owner of United States Letters Patent No. 7,504,095 and Market Applicant for NULIBRY (fosdenopterin) NDA number 214018, Sentynl hereby submits this Supplemental to the prior filed Application for Extension of Patent Term under 35 U.S.C. §156 on April 16, 2021. Information in accordance with the requirements specified in 37 C.F.R. § 1.740 is provided herein.

In the fourth paragraph in the INTRODUCTORY COMMENTS please replace “Origin Biosciences, Inc.” with “Sentynl Therapeutics, Inc.” and replace “hereinafter “Origin Biosciences” with “Sentynl” such that the paragraph will read -

Applicant represents that Sentynl Therapeutics, Inc. (hereinafter “Sentynl”) the assignee of the entire right, title and interest in and to U.S. Patent No. 7,504,095 (the “Patent”) by virtue of a series of assignments listed in detail in the table below and summarized in the chain of title abstract included herewith as Substitute **Exhibit 2**. Copies of the assignment and cover sheet as recorded are included herewith as exhibits as indicated in the below table.

Please replace Exhibit 2 with attached Substitute Exhibit 2 included herewith.

Please replace the chart following the fourth paragraph in the INTRODUCTORY COMMENTS to read –

Exhibit #	Conveying Party	Receiving Party	Reel/ Frame	Date recorded
3	Inventors: 1. Guenter Schwarz 2. Ralf Mendel	TECHNISCHE UNIVERSITAT BRAUNSCHWEIG	018624/ 0800	December 13, 2006

Exhibit #	Conveying Party	Receiving Party	Reel/ Frame	Date recorded
	3. Jose Santamaria 4. Jochen Reiss			
4	THE TECHNICAL UNIVERSITY OF BRAUNSCHWEIG (TECHNISCHE UNIVERSITÄT BRAUNSCHWEIG)	ORPHATEC PHARMACEUTICALS GMBH	025837/ 0873	February 18, 2011
5a	ORPHATEC PHARMACEUTICAL S GMBH	ALEXION PHARMA INTERNATIONAL SARL	025836/ 0001	February 18, 2011
5b	ORPHATEC PHARMACEUTICAL S GMBH (Corrective Assignment) ¹	ALEXION PHARMA INTERNATIONAL SARL	055851/ 0383	March 30, 2021
6	ALEXION PHARMA INTERNATIONAL SARL ²	ALEXION PHARMA GMBH	048284/ 0665	February 7, 2019
7	ALEXION PHARMA GMBH	ALEXION PHARMA HOLDING UNLIMITED COMPANY	048280/ 0369	February 8, 2019
8	ALEXION PHARMA HOLDING UNLIMITED COMPANY	ALEXION PHARMACEUTICALS, INC.	048292/ 0722	February 11, 2019
9	ALEXION PHARMACEUTICAL S, INC.	ORIGIN BIOSCIENCES, INC.	048664/ 0789	March 21, 2019
22	ORIGIN BIOSCIENCES, INC.	SENTYNL THERAPEUTICS, INC.	059671/ 0674	April 21, 2022

Attached Exhibit 22 is included herewith.

Please replace the paragraph immediately following the chart in the INTRODUCTORY COMMENTS to read –

¹ A corrective assignment was recorded on March 30, 2021, to correct a typographical error in the cover sheet (date of execution). The notice of recordation and underlying documents for recordation are included herewith as **Exhibit 5a**.

² Alexion Pharma International SARL underwent a name change to Alexion Pharma GMBH on August 17, 2015, as evidenced by the document recorded and included as **Exhibit 5b**.

Sentynl is the current Marketing Applicant (see **Exhibit 23**) and has acquired and accepted full ownership of NDA number 214018 from Origin Biosciences, who conducted certain studies leading to the approval by the FDA of NULIBRY (fosdenopterin) for injection, for intravenous use. Sentynl will commercialize the approved product. Sentynl represents that it is authorized to rely on activities undertaken before the FDA for obtaining marketing approval for the approved product, NULIBRY (fosdenopterin) for injection, for intravenous use. The activities before the FDA on which Sentynl is authorized to rely includes NDA number 214018 and studies conducted under IND number 117502, as described below.

Attached Exhibit 24 is included herewith.

Please replace the third paragraph following the chart in the INTRODUCTORY COMMENTS to read –

Marketing approval of the NULIBRY (fosdenopterin) for injection, for intravenous use was based on NDA number 214018, submitted by Origin Biosciences (see **Exhibits 10-12**), and includes data based on studies conducted under IND number 117502. On March 31, 2022, Original Biosciences, Inc. authorized the FDA to transfer NDA number 214018 to Sentynl (See **Exhibit 24**).

Attached Exhibit 24 is included herewith.

Please replace the first paragraph in Section 11 to read -

A brief description of the significant activities by the Marketing Applicant (Sentynl⁶) during the applicable regulatory review period with respect to the approved product and the dates applicable to these significant activities are set forth in a chronological listing of events in **Exhibit 21**.

Please replace footnote 6 to read -

Sentynl is the current Marketing Applicant for NDA number 214018. However, as detailed in the Introductory Comments section of this application, Sentynl was not the original Marketing Applicant of NDA 214018. The NDA number 214018 was transferred from Original Biosciences. Origin Biosciences was the prior sponsor of NDA number 214018 and IND number 117502. Sponsorship of the IND number 117502 was transferred from Alexion Pharma

International SARL to Alexion Pharmaceuticals, Inc. and then from Alexion Pharmaceuticals, Inc. to Origin Biosciences. Sentynl is authorized to rely on the activities undertaken by the prior sponsors of the IND and NDA.

Please replace Section 14 to read -

Payment of the fee prescribed in 37 C.F.R. §1.20(j)(1) for a patent term extension application under 35 U.S.C. §156 was authorized to be charged against **Deposit Account No. 50-1283** referencing docket no. **ORGB-001/00US 336516-2001**. The undersigned authorizes any additional required fees to be deducted from, or any overpayments to be credited to, **Deposit Account No. 033975** referencing docket no. **074540-0568897**.

Please replace Section 15 to read -

15. Name and Address for Correspondence [37 C.F.R. 1.740(a)(15)]

Michelle L. Mehok
Reg. No. 53,314
PILLSBURY WINTHROP SHAW PITTMAN LLP
11682 El Camino Real
Suite 200
San Diego, California 92130
Email: Docket_IP@pillsburylaw.com
Telephone: 858-509-4071
Facsimile: 858-509-4010

Please amend the INDEX OF EXHIBITS to replace Exhibit 2 with Substitute Exhibit 2 and add to the end Exhibits 22-24 –

Supplemental Exhibits

Substitute Exhibit 2: Abstract of Chain of Title from USPTO Assignment Database

Exhibit 22: Assignment Cover Sheet and Assignment from ORIGIN BIOSCIENCES, INC. to SENTYNL THERAPEUTICS, INC. (reel/frame 059671/0674)

Exhibit 23: March 31, 2022, letter from Sentynl Therapeutics, Inc. to FDA providing notice of the acquisition of full ownership of and all rights to NULIBRY NDA number 214018, attachments and acknowledgement of receipt from the FDA.

Exhibit 24: March 31, 2022, letter from Origin Biosciences, Inc. to FDA providing notice of the change of ownership of the NULIBRY NDA number 214018 to Sentynl Therapeutics, Inc. and acknowledgement of receipt from the FDA.

Substitute Exhibit 2



Assignment abstract of title for Application 11343489

Invention title/Inventor	Patent	Publication	Application	PCT	International registration
METHOD FOR	7504095	20070037250	11343489		
OBTAINING	Mar 17, 2009	Feb 15, 2007	Jan 31, 2006		
PRECURSOR Z AND USE					
THEREOF FOR THE					
PRODUCTION OF A					
MEANS FOR THERAPY					
OF HUMAN					
MOLYBDENUM					
COFACTOR DEFICIENCY					
Guenter Schwarz, Ralf					
Mendel, Jose					
Santamaria, Jochen					
Reiss					

Assignments (11 total)

Assignment 11

Reel/frame	Execution date	Date recorded	Properties	Pages
059671/0674	Mar 31, 2022	Apr 21, 2022	2	7
Conveyance				
ASSIGNMENT OF ASSIGNORS INTEREST (SEE DOCUMENT FOR DETAILS).				
Assignors		Correspondent		
ORIGIN BIOSCIENCES, INC.		PILLSBURY WINTHROP SHAW PITTMAN LLP (CV)		
		11682 EL CAMINO REAL, SUITE 200		
		SAN DIEGO, CA 92130		

Assignee
 SENTYNL THERAPEUTICS, INC.
 420 STEVENS AVENUE, SUITE 200
 SOLANA BEACH, CALIFORNIA 92075

Assignment 10

Reel/frame	Execution date	Date recorded	Properties	Pages
059577/0483	Mar 31, 2022	Apr 12, 2022	2	6
Conveyance				
RELEASE BY SECURED PARTY (SEE DOCUMENT FOR DETAILS).				
Assignors		Correspondent		
U.S. BANK NATIONAL ASSOCIATION, AS COLLATERAL AGENT		MICHELLE MEHOK		
		11682 EL CAMINO REAL, SUITE 200		
		SAN DIEGO, CA 92130		

Assignee
 ORIGIN BIOSCIENCES, INC.
 421 KIPLING STREET
 PALO ALTO, CALIFORNIA 94301

Assignment 9

Reel/frame	Execution date	Date recorded	Properties	Pages
058144/0302	Nov 17, 2021	Nov 17, 2021	19	12
Conveyance				
SECURITY INTEREST (SEE DOCUMENT FOR DETAILS).				
Assignors		Correspondent		
BRIDGEBIO PHARMA, INC.		PILLSBURY WINTHROP SHAW PITTMAN LLP		
ORIGIN BIOSCIENCES, INC.		12255 EL CAMINO REAL		
EIDOS THERAPEUTICS, INC.		SUITE 300		
QED THERAPEUTICS, INC.		SAN DIEGO, CA 92130-4088		
ADRENAS THERAPEUTICS, INC.				
PHOENIX TISSUE REPAIR, INC.				

Assignee
 U.S. BANK NATIONAL ASSOCIATION, AS COLLATERAL AGENT
 GLOBAL CORPORATE TRUST
 1 FEDERAL ST.
 BOSTON, MASSACHUSETTS 02110

Assignment 8

Reel/frame	Execution date	Date recorded	Properties	Pages
048664/0789	Mar 7, 2019	Mar 21, 2019	1	6
Conveyance				
ASSIGNMENT OF ASSIGNORS INTEREST (SEE DOCUMENT FOR DETAILS).				

Assignors
ALEXION PHARMACEUTICALS, INC.

Correspondent
DT WARD PC
142A MAIN STREET
GROTON, MA 01450

Assignee
ORIGIN BIOSCIENCES, INC.
421 KIPLING STREET
PALO ALTO, CALIFORNIA 94301

Assignment 7

Reel/frame 048292/0722	Execution date Feb 8, 2019	Date recorded Feb 11, 2019	Properties 1	Pages 3
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Conveyance
ASSIGNMENT OF ASSIGNORS INTEREST (SEE DOCUMENT FOR DETAILS).

Assignors
ALEXION PHARMA HOLDING UNLIMITED COMPANY

Correspondent
ALEXION PHARMACEUTICALS, INC.
100 COLLEGE STREET
NEW HAVEN, CT 06510

Assignee
ALEXION PHARMACEUTICALS, INC.
100 COLLEGE STREET
NEW HAVEN, CONNECTICUT 06510

Assignment 6

Reel/frame 048280/0369	Execution date Feb 7, 2019	Date recorded Feb 8, 2019	Properties 1	Pages 4
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Conveyance
ASSIGNMENT OF ASSIGNORS INTEREST (SEE DOCUMENT FOR DETAILS).

Assignors
ALEXION PHARMA GMBH

Correspondent
ALEXION PHARMACEUTICALS, INC.
100 COLLEGE STREET
NEW HAVEN, CT 06510

Assignee
ALEXION PHARMA HOLDING UNLIMITED COMPANY
CANON'S COURT 22 VICTORIA STREET
HAMILTON HM12
BERMUDA

Assignment 5

Reel/frame 048284/0665	Execution date Aug 17, 2015	Date recorded Feb 7, 2019	Properties 1	Pages 4
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Conveyance
CHANGE OF NAME (SEE DOCUMENT FOR DETAILS).

Assignors
ALEXION PHARMA INTERNATIONAL SARL

Correspondent
ALEXION PHARMACEUTICALS, INC.
100 COLLEGE STREET
NEW HAVEN, CT 06510

Assignee
ALEXION PHARMA GMBH
GIESHUBELSTRASSE 30
ZURICH 8045
SWITZERLAND

Assignment 4

Reel/frame 025836/0001	Execution date Feb 8, 2011	Date recorded Feb 18, 2011	Properties 1	Pages 8
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Conveyance
ASSIGNMENT OF ASSIGNORS INTEREST (SEE DOCUMENT FOR DETAILS).

Assignors
ORPHATEC PHARMACEUTICALS GMBH

Correspondent
ROPES & GRAY LLP
PRUDENTIAL TOWER 800 BOYLSTON STREET
BOSTON, MA 02199-3600

Assignee
ALEXION PHARMA INTERNATIONAL SARL
AVENUE DU TRIBUNAL FEDERAL 34
LAUSANNE 1005
SWITZERLAND

Assignment 3

Reel/frame 055851/0383	Execution date Feb 7, 2011	Date recorded Mar 30, 2021	Properties 1	Pages 10
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Conveyance

CORRECTIVE ASSIGNMENT TO CORRECT THE EXECUTION DATE PREVIOUSLY RECORDED AT REEL: 25836 FRAME: 001. ASSIGNOR(S) HEREBY CONFIRMS THE ASSIGNMENT.

Assignors
ORPHATEC PHARMACEUTICALS GMBH

Correspondent
COOLEY LLP
1299 PENNSYLVANIA AVENUE, NW
ATTN: IP DOCKETING DEPARTMENT
WASHINGTON, DC 20004-2400

Assignee
ALEXION PHARMA INTERNATIONAL SARL
AVENUE DU TRIBUNAL FEDERAL 34
LAUSANNE 1005
SWITZERLAND

Assignment 2

Reel/frame	Execution date	Date recorded	Properties	Pages
025837/0873	Feb 7, 2011	Feb 18, 2011	1	4

Conveyance
ASSIGNMENT OF ASSIGNORS INTEREST (SEE DOCUMENT FOR DETAILS).

Assignors
THE TECHNICAL UNIVERSITY OF BRAUNSCHWEIG

Correspondent
ROPES & GRAY LLP
PRUDENTIAL CENTER 800 BOYLSTON STREET
BOSTON, MA 02199-3600

Assignee
ORPHATEC PHARMACEUTICALS GMBH
VIKTORIA WEG 7
NIEDERKASSEL DE-53859
GERMANY

Assignment 1

Reel/frame	Execution date	Date recorded	Properties	Pages
018624/0800	Jul 31, 2006	Dec 13, 2006	1	7

Conveyance
ASSIGNMENT OF ASSIGNORS INTEREST (SEE DOCUMENT FOR DETAILS).

Assignors
SCHWARZ, GUENTER
MENDEL, RALF
SANTAMARIA, JOSE
REISS, JOCHEN

Correspondent
AKERMAN SENTERFITT
P.O. BOX 3188
WEST PALM BEACH, FL 33402-3188

Assignee
TECHNISCHE UNIVERSITAT BRAUNSCHWEIG
POCKELSTRASSE 14
BRAUNSCHWEIG D-38106
GERMANY

Exhibit 22

PATENT ASSIGNMENT COVER SHEET

Electronic Version v1.1
 Stylesheet Version v1.2

EPAS ID: PAT7292385

SUBMISSION TYPE:	NEW ASSIGNMENT	
NATURE OF CONVEYANCE:	ASSIGNMENT	
CONVEYING PARTY DATA		
	Name	Execution Date
	ORIGIN BIOSCIENCES, INC.	03/31/2022
RECEIVING PARTY DATA		
Name:	SENTYNL THERAPEUTICS, INC.	
Street Address:	420 STEVENS AVENUE, SUITE 200	
City:	SOLANA BEACH	
State/Country:	CALIFORNIA	
Postal Code:	92075	
PROPERTY NUMBERS Total: 2		
Property Type	Number	
Patent Number:	7504095	
Patent Number:	9260462	
CORRESPONDENCE DATA		
Fax Number:	(703)770-7901	
<i>Correspondence will be sent to the e-mail address first; if that is unsuccessful, it will be sent using a fax number, if provided; if that is unsuccessful, it will be sent via US Mail.</i>		
Phone:	8585094088	
Email:	docket_ip@pillsburylaw.com, michelle.mehok@pillsburylaw.com, sandra.edge@pillsburylaw.com	
Correspondent Name:	PILLSBURY WINTHROP SHAW PITTMAN LLP (CV)	
Address Line 1:	11682 EL CAMINO REAL, SUITE 200	
Address Line 4:	SAN DIEGO, CALIFORNIA 92130	
ATTORNEY DOCKET NUMBER:	074540-0000016	
NAME OF SUBMITTER:	MICHELLE MEHOK	
SIGNATURE:	/Michelle Mehok/	
DATE SIGNED:	04/21/2022	
Total Attachments: 5		
source=Sentynl - Nulibry - Patent Assignment Agreement#page1.tif		
source=Sentynl - Nulibry - Patent Assignment Agreement#page2.tif		
source=Sentynl - Nulibry - Patent Assignment Agreement#page3.tif		
source=Sentynl - Nulibry - Patent Assignment Agreement#page4.tif		

PATENT ASSIGNMENT AGREEMENT

This PATENT ASSIGNMENT AGREEMENT (this "Assignment"), effective as of March 31, 2022 (the "Effective Date"), is made by and between Origin Biosciences, Inc., a Delaware corporation ("Assignor") and Sentyln Therapeutics, Inc., a Delaware corporation ("Assignee"). Assignee and Assignor are sometimes herein each referred to as a "Party" and collectively as the "Parties." Capitalized terms used herein but not otherwise defined herein shall have the meanings set forth in the Agreement (defined below).

WHEREAS, Assignor and Assignee have entered into that certain Asset Purchase Agreement, dated as of March 4, 2022 (as may be amended or restated from time to time, the "Agreement"), pursuant to which, among other things, Assignor agreed to sell, assign, transfer and deliver to Assignee certain Intellectual Property (as defined in the Agreement), as more fully described in the Agreement, on the terms and subject to the conditions set forth in the Agreement;

WHEREAS, the Intellectual Property includes, without limitation, the Patent Rights held in, used by or otherwise related to or constituting the Business, including the Patent Rights set forth on Exhibit A hereto (collectively, the "Transferred Patents"); and

WHEREAS, Assignor and Assignee now desire to enter into this Assignment to evidence the sale, assignment, transfer and delivery to Assignee of the Transferred Patents;

NOW THEREFORE, for good and valuable consideration, including the consideration set forth in the Agreement, the receipt and sufficiency of which is hereby acknowledged, Assignor and Assignee hereby agree as follows:

1. Assignor hereby sells, assigns, transfers and delivers to Assignee, and Assignee hereby purchases and acquires from Assignor, all rights, title and interests of Assignor in and to the Transferred Patents, for the United States and for all foreign countries (including, without limitation, all rights therein provided by international conventions and treaties), including, without limitation, any reissues, divisions, continuations, continuations-in-part, extensions, re-examinations and foreign equivalents thereof, all inventions disclosed therein and improvements thereto, all rights to claim priority thereto or therefrom, and all rights therein provided by international treaties and conventions, together with all associated income, royalties, damages or payments accrued, due or payable as of the Effective Date or thereafter and including, without limitation, all claims for damages, restitution, and injunctive and other legal and equitable relief by reason of past, present or future infringement, misappropriation, violation, misuse, breach or other unauthorized use of the Transferred Patents, with the right to sue for, and collect the same, in each case, for Assignee's own use and enjoyment, and for the use and enjoyment of its successors, assigns or other legal representatives, as fully and entirely as the same would have been held and enjoyed by Assignor if this Assignment and sale had not been made.

2. Assignor hereby authorizes and requests the Commissioner of Patents of the United States, and any other official of any applicable Governmental Authority, to record this Assignment and to issue any and all registrations from any and all applications for registration included in the Transferred Patents to and in the name of Assignee.

3. Assignor shall, at the cost and expense of Assignee, timely execute and deliver any additional documents and perform such additional acts necessary or desirable to record and perfect the interest of Assignee in and to the Transferred Patents.

4. This Assignment is subject to the terms and conditions of the Agreement, which are incorporated by reference, and shall be binding upon Assignor and Assignee, and their respective successors and assigns.

5. This Assignment shall be governed by and construed and interpreted in accordance with the laws of the State of Delaware, irrespective of the choice of laws principles of Delaware, as to all matters, including matters of validity, construction, effect, enforceability, performance and remedies.

6. This Assignment may be executed in counterparts each of which shall be deemed an original and all of which, when taken together, shall constitute one and the same instrument.

7. This Assignment shall be binding upon and inure to the benefit of and be enforceable by the successors, legal representatives and permitted assigns of each party hereto.

[Remainder of page intentionally left blank]

Each of Assignor and Assignee has caused this Assignment to be executed by a duly authorized officer as of the date first written above.

ASSIGNOR

ORIGIN BIOSCIENCES, INC.

DocuSigned by:

By: _____
Name: Michael Henderson, M.D.
Title: Chief Executive Officer

[Signature Page to Patent Assignment Agreement]

Each of Assignor and Assignee has caused this Assignment to be executed by a duly authorized officer as of the date first written above.

SENTYNL THERAPEUTICS, INC.

By:  _____
DocuSigned by:
Name: Matthew T. Heck
Title: Chief Executive Officer

[Signature Page to Patent Assignment Agreement]

Exhibit A

Transferred Patents

Patent Ref.	Patent No.	Current Owner(s)	Jurisdiction	Issuance Date
Method for Obtaining Precursor Z and Use Thereof for the Production of a Means for Therapy of Human Molybdenum Cofactor Deficiency	7,504,095	Origin	US	17-Mar-2009
Method for Obtaining Precursor Z and Use Thereof for the Production of a Means for Therapy of Human Molybdenum Cofactor Deficiency	2,554,461	Origin	CA	22-Jan-2013
Method for Synthesizing Molybdopterin Precursor Z Derivatives	9,260,462	Origin	US	16-Feb-2016
Method for Synthesizing Molybdopterin Precursor Z Derivatives	2,827,679	Origin	CA	2-Jun-2020

Exhibit 23



420 Stevens Avenue, Suite 200
Solana Beach, CA 92075

March 31, 2022

Kathleen Donohue, M.D., Division Director
Food and Drug Administration
Center for Drug Evaluation and Research
Office of Rare Diseases, Pediatrics, Urologic
and Reproductive Medicine (ORPURM)
Division of Rare Diseases and Medical Genetics (DRDMG)
10903 New Hampshire Avenue
Silver Spring, MD 20993

RE: NDA 214018 eCTD seq 0073
NULIBRY™ (fosdenopterin) for Injection
General Correspondence – Change in Ownership per 21 CFR 314.72

Dear Dr. Donohue,

Reference is made to the Origin Biosciences, Inc. (Origin) New Drug Application (NDA) 214018 for Nulibry (fosdenopterin) for Injection, indicated to reduce the risk of mortality in patients with molybdenum cofactor deficiency (MoCD) Type A, approved on 26 February 2021.

In accordance with 21 CFR 314.72, Sentyln Therapeutics, Inc. ("Sentyln") hereby notifies the U.S. Food and Drug Administration that it has acquired and accepted full ownership of and all rights to the above-referenced NDA from Origin, the former owner of the NDA.

The effective date of the change in ownership and transfer of the NDA is March 31, 2022. Sentyln commits to all agreements, promises and conditions made by Origin and contained in the NDA. Sentyln has a complete copy of the NDA, including supplements and records that are required to be kept under 21 CFR 314.81 and is not requesting a copy of the NDA from the FDA's files. Sentyln agrees to advise FDA of any change in the conditions in the approved application pursuant to 21 CFR 314.70.

A copy of Origin's letter notifying the Agency of this transfer of ownership of the NDA to Sentyln is provided with this submission and will be provided in original form to FDA under separate cover by Origin (NDA 214018; eCTD seq 0072).

This submission includes a completed copy of Form FDA 356h and updates to Module 1.3.1.

This submission has been prepared in electronic Common Technical Document (eCTD) format and submitted by Synteract through the Agency's Electronic Submissions Gateway. Synteract, Inc. certifies that precautions have been taken to ensure that the eCTD is free of computer viruses. If you have any information technology (IT) questions regarding this submission, please do not hesitate to contact Thomas Christensen by phone at (919) 674-8099 or by email at Thomas.Christensen@synteract.com.



420 Stevens Avenue, Suite 200
Solana Beach, CA 92075

Should you have any questions regarding this submission, please feel free to contact me by telephone at 858-314-4223 or by email at ebanaga@sentynl.com.

Sincerely,

Eileen Banaga

Digitally signed by Eileen
Banaga
Date: 2022.03.31 08:48:49 -07'00'

Eileen Banaga, MS, RAC
Sr. Director, Regulatory Affairs and Quality
Sentynl Therapeutics, Inc.
420 Stevens Ave., Suite 200
Solana Beach, CA 92075
ebanaga@sentynl.com
858-314-4223 (Office)



31 March 2022

Kathleen Donohue, M.D., Division Director
Food and Drug Administration
Center for Drug Evaluation and Research
Office of Rare Diseases, Pediatrics, Urologic and Reproductive Medicine (ORPURM)
Division of Rare Diseases and Medical Genetics (DRDMG)
10903 New Hampshire Avenue
Silver Spring, MD 20993

**RE: NDA 214018; Sequence Number (Seq)-0072
NULIBRY™ (fosdenopterin) for Injection for the Treatment of MoCD Type A
Product Correspondence: Change of Ownership of an Application**

Dear Dr. Donohue,

Reference is made to the Origin Biosciences, Inc. (Origin) New Drug Application (NDA) number 214018 for Nulibry (fosdenopterin) for Injection as a treatment of Molybdenum Cofactor Deficiency (MoCD) Type A approved on 26 February 2021.

In accordance with §21 CFR 314.72, Origin is herewith notifying FDA of a change of ownership of the Nulibry NDA 214018 from Origin Biosciences, Inc. to Sentyln Therapeutics, Inc. All rights to this NDA have been transferred to Sentyln Therapeutics, Inc. effective immediately.

This change of ownership will be confirmed in a subsequent submission, Seq-0073 from the new owner, Sentyln Therapeutics, Inc. Until completion of the Sentyln Therapeutics, Inc. submission, preliminary contact information for Sentyln Therapeutics, Inc. is as follows:

Ms. Eileen Banaga
Sr. Director, Regulatory Affairs and Quality
Sentyln Therapeutics, Inc.
420 Stevens Ave., Suite 200
Solana Beach, CA 92075
858-314-4223 (Office)
ebanaga@sentyln.com

This submission is provided electronically in Common Technical Document (eCTD) format and is being transmitted through the Electronic Submission Gateway. The files contained in the submission were scanned for viruses using the current version of Sophos Endpoint Security and Control (version 10.8.10.3 Sophos). No viruses were detected, and the Applicant certifies that the submission is virus-free. For issues/problems associated with the technical processing of this

electronic submission, please contact Chris Rafael at 619-370-8599 or chris.rafael@bridgebio.com.

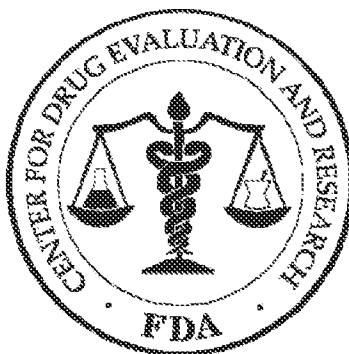
Should you have any questions regarding this submission, please contact me via email (jpavillard@origintx.com) or by phone at (610) 716-6391. If I am unavailable, please contact Kyna Williams via email at kwilliams@origintx.com.

Sincerely,
DocuSigned by Jennifer Pavillard

 *Jennifer Pavillard* | I am the author of this document
3/30/2022 | 1:22 PDT
E546BEDAEF6C4094AAB51109E0E57E9B

Jennifer Pavillard
Vice President, Regulatory Affairs
Origin Biosciences, Inc.

Trade secret and/or confidential commercial information contained in this submission is exempt from public disclosure to the full extent provided by/under law.



Your submission was successfully processed into the CDER Electronic Document Room, and is available to the assigned review division.

Application Type/Number:	NDA214018
eCTD Sequence Number:	0072
CoreID:	ci1648764368224.5082852@fdslv86001_te1

Your official receipt date is calculated in accordance with the following final Guidance for Industry:
<https://www.fda.gov/regulatory-information/search-fda-guidance-documents/providing-regulatory-submissions-electronic-format-receipt-date>

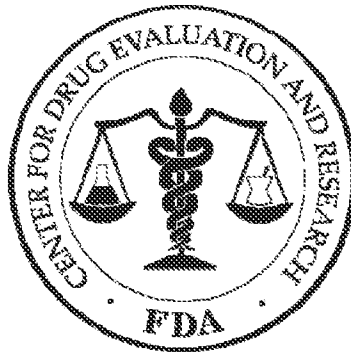
Contact Information:

For technical assistance only: eSUB@fda.hhs.gov

For all other questions regarding your submission, contact your review division.

Thank you,

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Center for Drug Evaluation and Research
U.S. Food and Drug Administration



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Application Type/Number:	NDA214018
eCTD Sequence Number:	0073
CoreID:	ci1648769179102.8625571@fdslv86003_te1

Your official receipt date is calculated in accordance with the following final Guidance for Industry:
<https://www.fda.gov/regulatory-information/search-fda-guidance-documents/providing-regulatory-submissions-electronic-format-receipt-date>

Contact Information:

For technical assistance only: eSUB@fda.hhs.gov

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Thank you,

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Center for Drug Evaluation and Research
U.S. Food and Drug Administration

Exhibit 24



31 March 2022

Kathleen Donohue, M.D., Division Director
Food and Drug Administration
Center for Drug Evaluation and Research
Office of Rare Diseases, Pediatrics, Urologic and Reproductive Medicine (ORPURM)
Division of Rare Diseases and Medical Genetics (DRDMG)
10903 New Hampshire Avenue
Silver Spring, MD 20993

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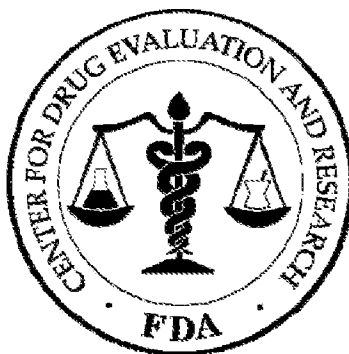
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